TM Forum Specification

Federated ID

TMF691

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Table of Contents

NOTICE . . . i
Introduction . . . 1
Sample Use Cases . . . 3
Use Case 1: An application needs to identify an end-user . . . 3
Description . . . 3
Use Case 2: An application needs to identify an end-user and fetch their permissions and manageable assets. . . . 3
Description . . . 3
Support of polymorphism and extension patterns . . . 4
RESOURCE MODEL . . . 5
Managed Entity and Task Resource Models . . . 5
Userinfo resource . . . 5
Notification Resource Models . . . 23
API OPERATIONS . . . 25
Operations on Userinfo . . . 25
Retrieves a Userinfo by ID . . . 25
List or find Userinfo objects . . . 27
API NOTIFICATIONS . . . 30
Register listener . . . 30
Unregister listener . . . 31
Publish Event to listener . . . 31
Acknowledgements . . . 33
Release History . . . 33
Contributors to Document . . . 33

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Introduction

The following document is the specification of the REST API for Federated ID Management. It includes the model definition as well as all available operations for SID userinfo entity. This API covers the operations required to allow an application (for instance a selfcare mobile app) request identity information about the individual that is making use of the functionality provided by such application (the user), or in general to allow an application to request identity related information about an individual to the system holding such identity information. This API, instead of defining new operations, relies on the use of industry standard for identity information such as OpenID Connect (http://openid.net/specs/openid-connect-core-1_0.html) which is a simple identity layer on top of the OAuth 2.0 (RFC6749) protocol.

OIDC enables Clients (for instance a selfcare mobile app) to verify the identity of the End-User based on the authentication performed by an Authorization Server, as well as to obtain basic profile information about the end user at the userinfo endpoint. Clients send requests to the Userinfo endpoint to obtain Claims about the end-user using an Access Token obtained through OpenID Connect Authentication. The Userinfo Endpoint is an OAuth 2.0 [RFC6749] Protected Resource that complies with the OAuth 2.0 Bearer Token Usage [RFC6750] specification and returns claims about the authenticated end user. The request uses the HTTP GET method and the access token is sent using the Authorization header. The following is a non-normative example of a UserInfo Request:

GET /userinfo HTTP/1.1
Host: server.example.com
Authorization: Bearer SlAV32hkKG46DAGFIK

The Federated Identity API therefore is a wrapper API that exposes the UserInfo endpoint protected resource that returns the claims about an authenticated end-user. This API manages Federated Identity because it defines the operations that must be supported by any system in order to allow providing identity related information (i.e.: the set of attributes related to the individual, such as name, family name, primary phone number, gender, birthdate, …) to different requesting applications, provided they are authorized to perform such request. Authorization can be granted by providing a valid proof of authorization (e.g.: an OAuth2.0 access token) granted by the individual whose identity is being requested.

It is up to each implementation to define how authorization can be provided (and confirmed), this API only defines the operations required to request identity related information by an authorized application. As indicated before industry standard mechanisms (OpenID Connect and OAuth2.0) are considered for the implementation of this API, therefore it is assumed in this specification that the request for user´s identity includes an Authorization header with a valid token.

```
Due to the possibility of token substitution attacks, the Userinfo response is not
guaranteed to be about the end-user identified by the sub (subject) element of the
OpenID Token. The implememtation MUST ensure that the sub Claim in the Userinfo
response is verified to exactly match the sub Claim in the ID Token; if they do not
match, the Userinfo response values MUST not be used.
```

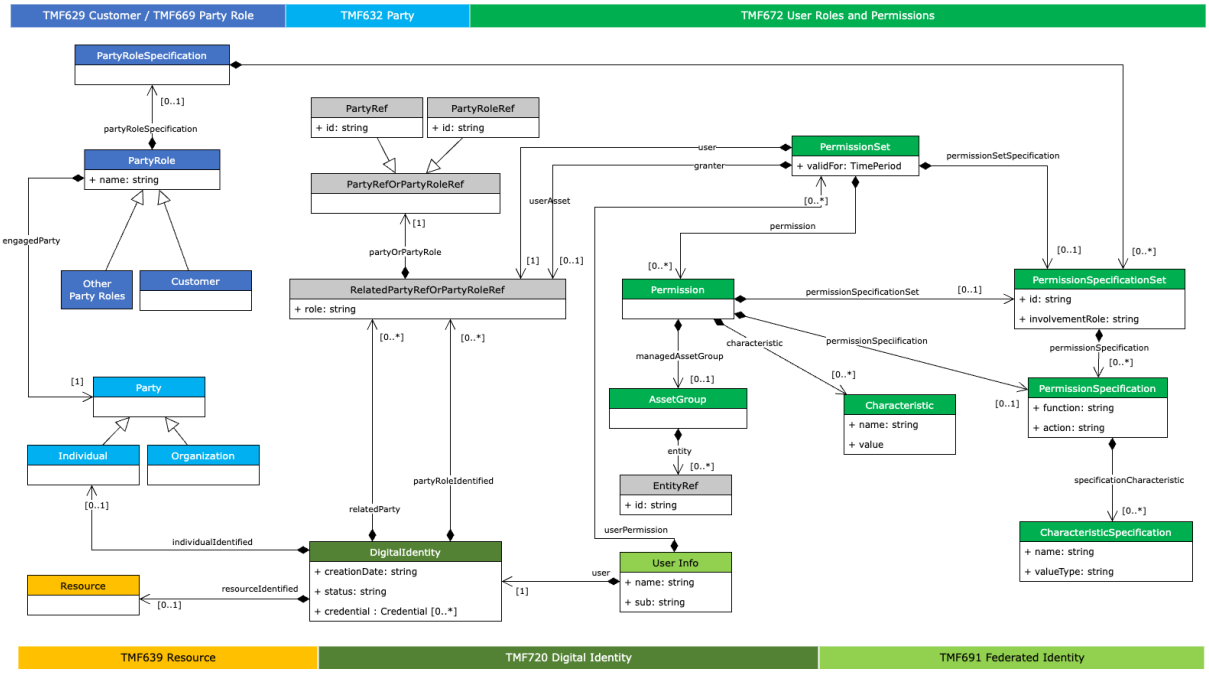
This API relates with existing TM Forum PartyManagement API (TMF632) because a user is actually an individual (i.e.: a party) acting with the role of “user” when interacting with the

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application requesting the identity information, therefore the so called “userinfo” can be actually either an extension or a subset of the specific party data information, which typically encompasses identity related and any other additional information related to such individual (such as for instance the date when the individual registered in the system, the status or the list of other related parties which is not identity-specific information).

This API defines a set of extended claims over those standard claims defined in OpenID Connect, in order to provide not only the basic user identity information but also the list of permissions on assets that is available to and can be managed by the individual whose identity information is being requested.

Following diagram illustrates the involved entities in a global party domain overview, referencing related TM Forum Open APIs and resource model entities:



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Sample Use Cases

This section includes a set of main use cases that can be performed with this API. Additional use cases can be generated using the operations and resources defined in this specification.

Use Case 1: An application needs to identify an end-user

Description

An individual is using a selfcare application. The application needs to provide functionality to the individual end user based on their identity. Therefore the application needs to identify the individual. In order to obtain the identity information of the individual, the application performs the following steps:

1. The application obtains a proof of authorization by the individual to request his identity information (e.g.: an OAUth2.0 token)
2. The application requests the identity-related information of the user (i.e.: the userinfo) associated to the individual whose proof of authorization is provided

Use Case 2: An application needs to identify an end-user and fetch their permissions and manageable assets.

Description

An individual is using a selfcare application. The application needs to provide functionality to the individual end user based on the information stored in a system e.g. billing information about the billing accounts associated to the user. Therefore the application needs to identify the individual and know what assets that user can manage and their persmission to those assets. In order to obtain the identity information of the individual, the application performs the following steps:

1. The application obtains a proof of authorization by the individual to request his identity information (e.g.: an OAUth2.0 token)
2. The application requests the identity-related information of the user (i.e.: the userinfo and a reference to the set of permissions) associated to the individual whose proof of authorization is provided

Reader will find additional example of use cases using Usage API in “Open Digital Business Scenarios and Use Cases” document.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

Userinfo resource

Resource model

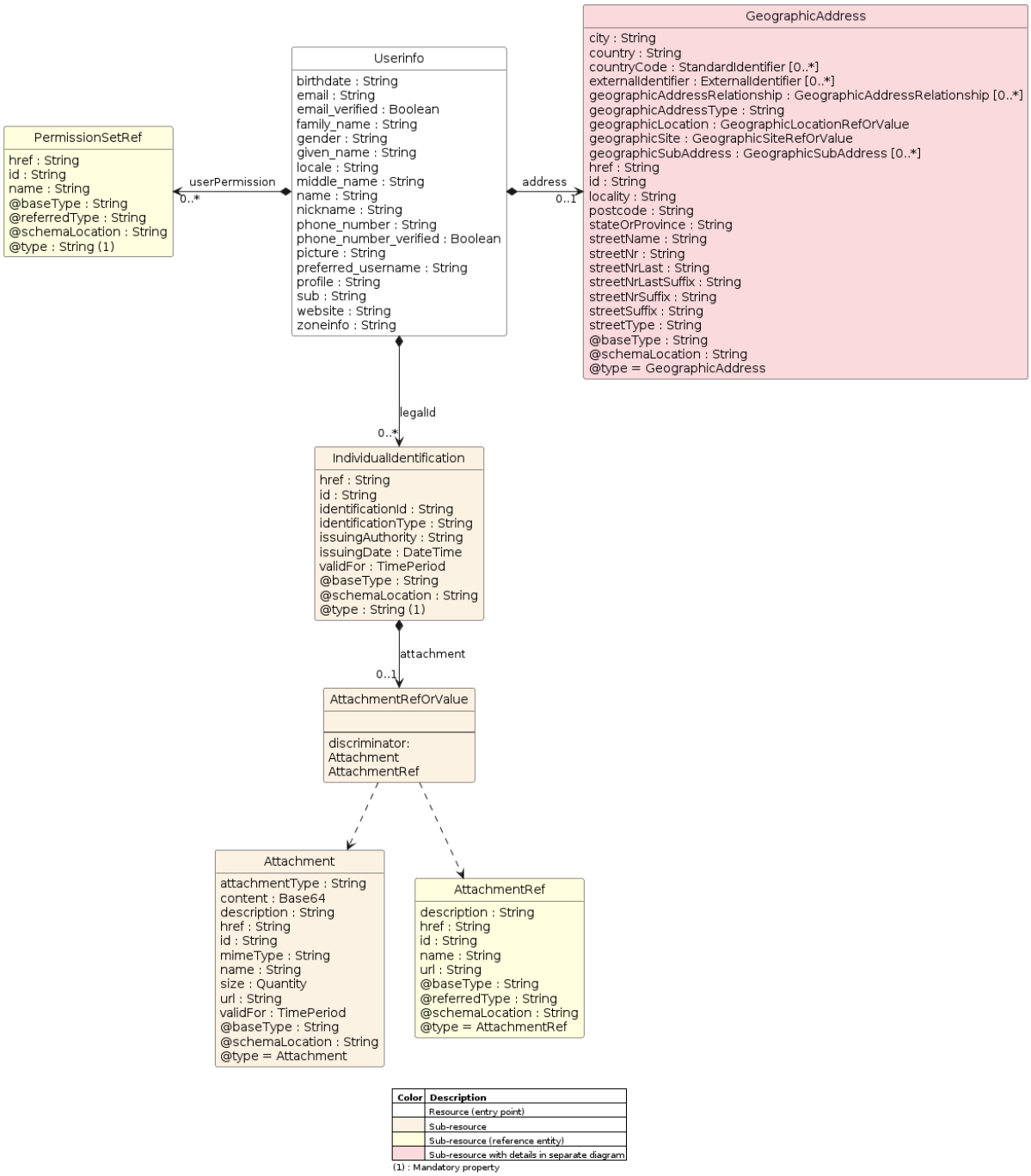


Figure 1 - Userinfo

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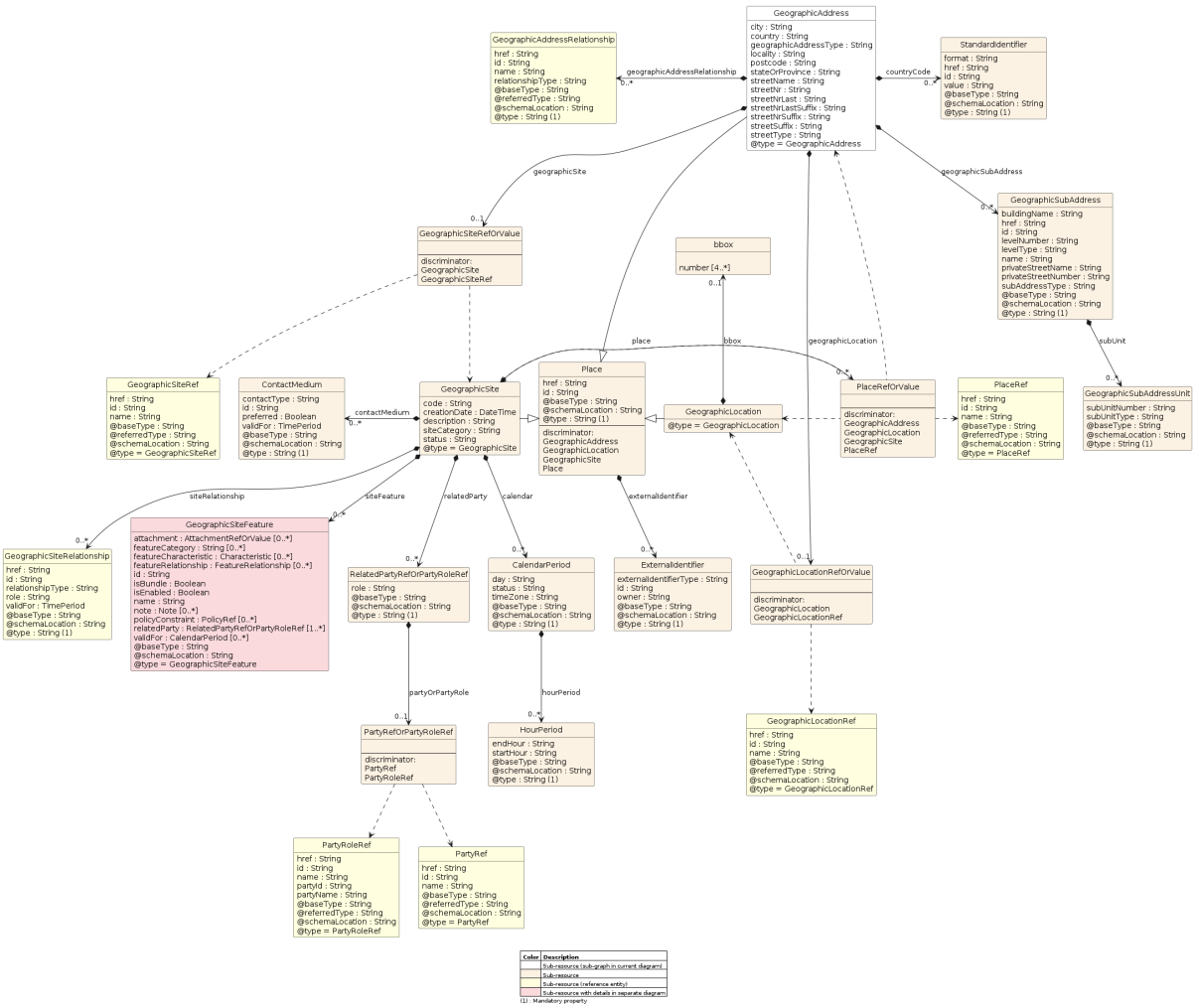


Figure 2 - GeographicAddress

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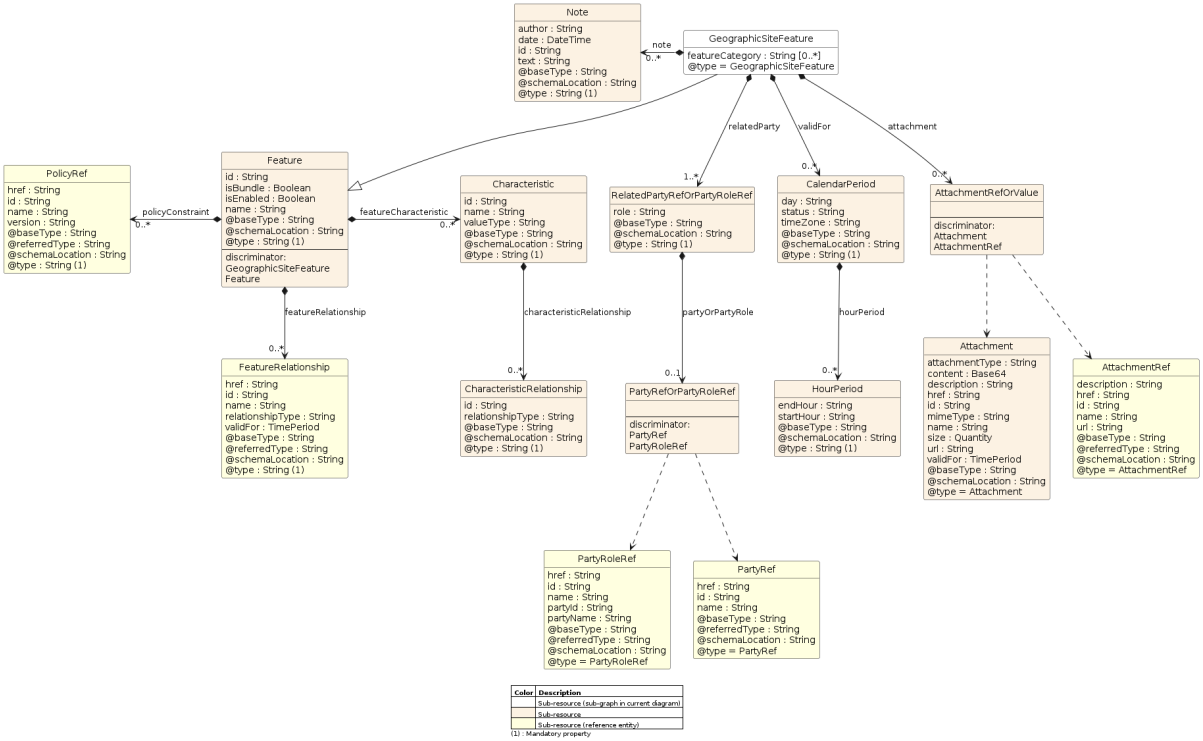


Figure 3 - GeographicSiteFeature

Field descriptions

Userinfo fields

address	A GeographicAddress. Structured textual way of describing how to find a Property in an urban area (country properties are often defined differently). Note : Address corresponds to SID UrbanPropertyAddress.
birthdate	A String. End-User's birthday, represented as an [ISO8601-2004] YYYY-MM-DD format. The year MAY be 0000, indicating that it is omitted. To represent only the year, YYYY format is allowed. Note that depending on the underlying platform's date related function, providing just year can result in varying month and day, so the implementers need to take this factor into account to correctly process the dates.
email	A String. End-User's preferred e-mail address. Its value MUST conform to the [RFC5322] addr-spec syntax.
email_verified	A Boolean. True if the user's email has been verified.
family_name	A String. Surname(s) or last name(s) of the End-User. Note that in some cultures, people can have multiple family names or no family name; all can be present, with the names being separated by space characters.
gender	A String. End-User's gender. Values defined by this specification are female and male. Other values MAY be used when neither of the defined values are applicable.

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given_name	A String. Given name(s) or first name(s) of the End-User. Note that in some cultures, people can have multiple given names; all can be present, with the names being separated by space characters.
legalId	An IndividualIdentification. Identification documentation of the contact.
locale	A String. End-User's locale, represented as a [RFC5646] language tag. This is typically an [ISO639-1] language code in lowercase and an [ISO3166-1] country code in uppercase, separated by a dash. For example, en-US or fr-CA. As a compatibility note, some implementations have used an underscore as the separator rather than a dash, for example, en_US.
middle_name	A String. Middle name(s) of the End-User. Note that in some cultures, people can have multiple middle names; all can be present, with the names being separated by space characters. Also note that in some cultures, middle names are not used.
name	A String. End-User's full name in displayable form including all name parts, possibly including titles and suffixes, ordered according to the End-User's locale and preferences.
nickname	A String. Casual name of the End-User that may or may not be the same as the given_name. For instance, a nickname value of Mike might be returned alongside a given_name value of Michael.
phone_number	A String. End-User's preferred telephone number. [E.164] is RECOMMENDED as the format of this Claim, for example, +1 (425) 555-1212 or +56 82) 687 2400. If the phone number contains an extension, it is RECOMMENDED that the extension be represented using the [RFC3966] extension syntax, for example, +1 (604) 555-1234;ext=5678.
phone_number_verified	A Boolean. True if the user's phone number has been verified.
picture	A String. URL of the End-User's profile picture. This URL MUST refer to an image file (for example, a PNG, JPEG, or GIF image file), rather than to a Web page containing an image. Note that this URL SHOULD specifically reference a profile photo of the End-User suitable for displaying when describing the End-User, rather than an arbitrary photo taken by the End-User.
preferred_username	A String. Shorthand name by which the End-User wishes to be referred to at the RP, such as janeDoe or j.doe. This value MAY be any valid JSON string including special characters such as @, /, or whitespace.
profile	A String. URL of the End-User's profile page. The contents of this Web page SHOULD be about the End-User.
sub	A String. Subject - Unique Identifier for the End-User.
userPermission	A PermissionSetRef. The set of Permissions granted to the user (party, party role or resource).

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website	A String. URL of the End-User's Web page or blog. This Web page SHOULD contain information published by the End-User or an organization that the End-User is affiliated with.
zoneinfo	A String. String from 'zoneinfo' time zone database representing the End-User's time zone. For example, Europe/Paris or America/Los_Angeles.

PermissionSetRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddress sub-resource fields

city	A String. City that the address is in.
country	A String. Country that the address is in.
countryCode	A StandardIdentifier. The corresponding identification of the resource in different standard, regulatory definitions. The standard specification identifier (e.g., ISO 3166-1 Alpha-2) and the corresponding value (e.g., BE) relevant to a particular resource. It is anticipated that multiple standards can provide definitions for a single entity, e.g., a country identifier can be specified in various standards (e.g., "ISO 3166-1 Alpha 2", "ISO 3166-1 Alpha 3", "ISO 3166-1 Numeric").
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
geographicAddressRelationship	A GeographicAddressRelationship. The GeographicAddressRelationship schema represents a relationship between geographic addresses. It defines the structure for storing information about how two geographic addresses are related to each other within a system.

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geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSite	A GeographicSiteRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicSite entity and not the GeographicSiteRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress. Representation of a GeographicSubAddress It is used for addressing within a property in an urban area (country properties are often defined differently). It may refer to a building, a building cluster, or a floor of a multistory building.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

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bbox	A bbox. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

calendar	A CalendarPeriod. The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.
code	A String. A code that may be used for some addressing schemes eg: [ANSI T1.253-1999].
contactMedium	A ContactMedium. Indicates the contact medium that could be used to contact the party. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : EmailContactMedium, FaxContactMedium, PhoneContactMedium, GeographicAddressContactMedium, SocialMediaContactMedium…
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Text describing additional information regarding the site.

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externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
siteCategory	A String. Site classification/category.
siteFeature	A GeographicSiteFeature. Geographic Site Feature captures various site information, ranging from survey data to safety guidelines and hazard information.
siteRelationship	A GeographicSiteRelationship. Details of geographic site relationship.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Place sub-resource fields

externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Place can be instanciated as * GeographicAddress * GeographicLocation * GeographicSite
bbox	This property is present in subclasses
calendar	This property is present in subclasses
city	This property is present in subclasses
code	This property is present in subclasses
contactMedium	This property is present in subclasses
country	This property is present in subclasses
countryCode	This property is present in subclasses
creationDate	This property is present in subclasses
description	This property is present in subclasses
geographicAddressRelationship	This property is present in subclasses
geographicAddressType	This property is present in subclasses
geographicLocation	This property is present in subclasses
geographicSite	This property is present in subclasses
geographicSubAddress	This property is present in subclasses
locality	This property is present in subclasses
place	This property is present in subclasses
postcode	This property is present in subclasses
relatedParty	This property is present in subclasses
siteCategory	This property is present in subclasses
siteFeature	This property is present in subclasses
siteRelationship	This property is present in subclasses
stateOrProvince	This property is present in subclasses
status	This property is present in subclasses
streetName	This property is present in subclasses
streetNr	This property is present in subclasses
streetNrLast	This property is present in subclasses
streetNrLastSuffix	This property is present in subclasses

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streetNrSuffix	This property is present in subclasses
streetSuffix	This property is present in subclasses
streetType	This property is present in subclasses

IndividualIdentification sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
identificationId	A String. Identifier.
identificationType	A String. Identification type (passport, national identity card, drivers license, social security number, birth certificate).
issuingAuthority	A String. Authority which has issued the identifier, such as: social security, town hall.
issuingDate	A DateTime. Date at which the identifier was issued.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddressRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. Type of relationship between the geographic addresses. This attribute specifies the nature of the relationship between the related addresses. It provides context for understanding how the addresses are related to each other.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

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format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.

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subUnit	A GeographicSubAddressUnit. Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

subUnitNumber	A String. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ContactMedium sub-resource fields

contactType	A String. Type of the contact medium to qualify it like pro email / personal email. This is not used to define the contact medium used.
id	A String. Identifier for this contact medium.
preferred	A Boolean. If true, indicates that is the preferred contact medium.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A String. Day where the calendar status applies (e.g.: monday, mon-to-fri, weekdays, weekend, all week, …).
hourPeriod	A HourPeriod. Hour interval.
status	A String. Indication of the availability of the calendar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information (e.g.: Paris, GMT+1).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRelationship sub-resource fields

href	A String. Reference of the related geographic site.
id	A String. Unique identifier of the related site entity within the server.
relationshipType	A String. Type of relationship.
role	A String. Role of the related site in the relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteFeature sub-resource fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
featureCategory	A String. Collection of feature category.
featureCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
featureRelationship	A FeatureRelationship. Configuration feature.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
note	A Note. Extra information about a given entity.
policyConstraint	A PolicyRef. Reference to managed Policy object.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

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validFor	A CalendarPeriod. The CalendarPeriod schema represents a period of time within a calendar, defining various attributes such as the applicable day, timezone, hour period, and status. It is designed to capture information about the availability or status of a calendar period, which can be used in scheduling, booking, or resource allocation applications.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

featureCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
featureRelationship	A FeatureRelationship. Configuration feature.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to managed Policy object.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Feature can be instanciated as * GeographicSiteFeature
attachment	This property is present in subclasses
featureCategory	This property is present in subclasses
note	This property is present in subclasses
relatedParty	This property is present in subclasses
validFor	This property is present in subclasses

HourPeriod sub-resource fields

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endHour	A String. The time when the status ends applying.
startHour	A String. The time when the status starts applying.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.

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id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.

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@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'Userinfo' resource object.


```
{
    "id": "a066ffd5-f685-43bc-ad57-bfb7639decb0",
    "href": "https://host:port/userInfo/a066ffd5-f685-43bc-ad57-bfb7639decb0",
    "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Common/Userinfo.schema.json",
    "@type": "Userinfo",
    "address": {
      "id": "1a2b3c",
      "href": "https://host:port/location/geographicAddress/1a2b3c",
      "streetNr": "1",
      "streetName": "UnterDenLinden",
      "city": "Berlin",
      "country": "Germany",
      "postcode": "10117",
      "@type": "GeographicAddress",
      "geographicLocation":
      {
        "id": "afuih78g",
        "@type": "GeographicLocation"
      }
    },
    "birthdate": "1970-11-02",
    "email": "jane.doe@email.com",
    "email_verified": true,
    "family_name": "Doe",
    "gender": "female",
    "given_name": "Jane",
    "legalId": [
      {
        "@type" : "IndividualIdentification",
        "identificationType": "passport",
        "identificationId": "51DD1234",
        "issuingAuthority": "US Government",
        "issuingDate": "2017-10-22T07:49:25.246Z",
        "validFor": {
          "startDateTime": "2017-10-22T07:49:25.246Z",
          "endDateTime": "2027-10-21T07:49:25.246Z"
        },
        "attachment": [
          {
            "attachmentType": "passportCopy",
            "name": "PassportCopy321.pdf",
            "description": "Scanned copy of the passport",
            "mimeType": "application/pdf",
            "url": "http://DocumentManager/321/PassportCopy321.pdf",
            "@type": "Attachment",
            "id": "1003"
```


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```
          }
        ]
      }
    ],
    "userPermission":[
        {
          "id": "1492026921",
          "name": "End-user subscription permission set",
          "href": "https://mycsp.com/tmf-
api/rolesAndPermissionsManagement/v5/PermissionSet/1145",
          "@type": "PermissionSetRef",
          "@referredType": "PermissionSet"
        },
        {
          "id": "5753026921",
          "name": "End-user subscription permission set",
          "href": "https://mycsp.com/tmf-
api/rolesAndPermissionsManagement/v5/PermissionSet/5435648",
          "@type": "PermissionSetRef",
          "@referredType": "PermissionSet"
        }
    ],
    "locale": "en",
    "middle_name": "Mary",
    "name": "Jane Mary Doe",
    "nickname": "JaneDoe",
    "phone_number": "+1 (425) 555-1212",
    "phone_number_verified": true,
    "picture": "https://someurl.com/janeDoe.jpg",
    "preferred_username": "janeDoe",
    "profile": "https://my.blog.com/janeDoe",
    "sub": "a066ffd5-f685-43bc-ad57-bfb7639decb0",
    "website": "https://janedoe.com",
    "zoneinfo": "America/Los_Angeles"
  }
```


Notification Resource Models

0 notifications are defined for this API.

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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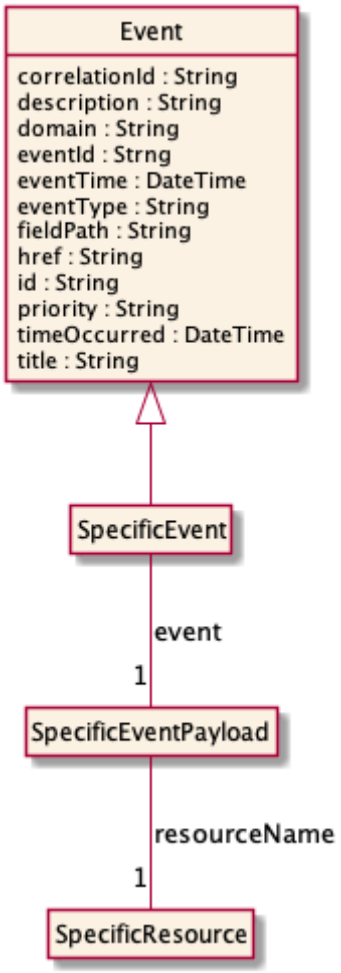


Figure 4 Notification Pattern

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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on Userinfo

Retrieves a Userinfo by ID

`GET /userinfo/{id}?fields=…`

Description

This operation retrieves a Userinfo entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

The Userinfo entity is not a list and therefore there is no operation to retrive userinfo by ID. Userinfo is retrieved for the user specified in the header token.

Request

```
GET /userinfo/3180
Content-Type: application/json
```

Response

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```
200

{
    "id": "3180",
    "href": "https://host:port/userInfo/a066ffd5-f685-43bc-ad57-bfb7639decb0",
    "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Common/Userinfo.schema.json",
    "@type": "Userinfo",
    "address": {
      "id": "1a2b3c",
      "href": "https://host:port/location/geographicAddress/1a2b3c",
      "streetNr": "1",
      "streetName": "UnterDenLinden",
      "city": "Berlin",
      "country": "Germany",
      "postcode": "10117",
      "@type": "GeographicAddress",
      "geographicLocation":
      {
        "id": "afuih78g",
        "@type": "GeographicLocation"
      }
    },
    "birthdate": "1970-11-02",
    "email": "jane.doe@email.com",
    "email_verified": true,
    "family_name": "Doe",
    "gender": "female",
    "given_name": "Jane",
    "legalId": [
      {
        "@type" : "IndividualIdentification",
        "identificationType": "passport",
        "identificationId": "51DD1234",
        "issuingAuthority": "US Government",
        "issuingDate": "2017-10-22T07:49:25.246Z",
        "validFor": {
          "startDateTime": "2017-10-22T07:49:25.246Z",
          "endDateTime": "2027-10-21T07:49:25.246Z"
        },
        "attachment": [
          {
            "attachmentType": "passportCopy",
            "name": "PassportCopy321.pdf",
            "description": "Scanned copy of the passport",
            "mimeType": "application/pdf",
            "url": "http://DocumentManager/321/PassportCopy321.pdf",
            "@type": "Attachment",
            "id": "1003"
          }
        ]
      }
    ],
    "userPermission":[
        {
          "id": "1492026921",
          "name": "End-user subscription permission set",
          "href": "https://mycsp.com/tmf-
api/rolesAndPermissionsManagement/v5/PermissionSet/1145",
          "@type": "PermissionSetRef",
          "@referredType": "PermissionSet"
        },
```

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```
        {
          "id": "5753026921",
          "name": "End-user subscription permission set",
          "href": "https://mycsp.com/tmf-
api/rolesAndPermissionsManagement/v5/PermissionSet/5435648",
          "@type": "PermissionSetRef",
          "@referredType": "PermissionSet"
        }
    ],
    "locale": "en",
    "middle_name": "Mary",
    "name": "Jane Mary Doe",
    "nickname": "JaneDoe",
    "phone_number": "+1 (425) 555-1212",
    "phone_number_verified": true,
    "picture": "https://someurl.com/janeDoe.jpg",
    "preferred_username": "janeDoe",
    "profile": "https://my.blog.com/janeDoe",
    "sub": "a066ffd5-f685-43bc-ad57-bfb7639decb0",
    "website": "https://janedoe.com",
    "zoneinfo": "America/Los_Angeles"
}
```


List or find Userinfo objects

`GET /userinfo?fields=…`

Description

This operation list Userinfo entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

This example details how userinfo is retrieved for a user. In this example a valid Authorization token must be present in the header.",

Request

```
GET /userinfo
Content-Type: application/json
```

Response


```
200

[
  {
  "id": "3180",
  "href": "https://host:port/userInfo/a066ffd5-f685-43bc-ad57-bfb7639decb0",
  "@schemaLocation": "https://mycsp.com:8080/tmf-
api/schema/Common/Userinfo.schema.json",
  "@type": "Userinfo",
  "address": {
```


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```
    "id": "1a2b3c",
    "href": "https://host:port/location/geographicAddress/1a2b3c",
    "streetNr": "1",
    "streetName": "UnterDenLinden",
    "city": "Berlin",
    "country": "Germany",
    "postcode": "10117",
    "@type": "GeographicAddress",
    "geographicLocation":
    {
      "id": "afuih78g",
      "@type": "GeographicLocation"
    }
  },
  "birthdate": "1970-11-02",
  "email": "jane.doe@email.com",
  "email_verified": true,
  "family_name": "Doe",
  "gender": "female",
  "given_name": "Jane",
  "legalId": [
    {
      "@type" : "IndividualIdentification",
      "identificationType": "passport",
      "identificationId": "51DD1234",
      "issuingAuthority": "US Government",
      "issuingDate": "2017-10-22T07:49:25.246Z",
      "validFor": {
        "startDateTime": "2017-10-22T07:49:25.246Z",
        "endDateTime": "2027-10-21T07:49:25.246Z"
      },
      "attachment": [
        {
          "attachmentType": "passportCopy",
          "name": "PassportCopy321.pdf",
          "description": "Scanned copy of the passport",
          "mimeType": "application/pdf",
          "url": "http://DocumentManager/321/PassportCopy321.pdf",
          "@type": "Attachment",
          "id": "1003"
        }
      ]
    }
  ],
  "userPermission":[
      {
        "id": "1492026921",
        "name": "End-user subscription permission set",
        "href": "https://mycsp.com/tmf-
api/rolesAndPermissionsManagement/v5/PermissionSet/1145",
        "@type": "PermissionSetRef",
        "@referredType": "PermissionSet"
      },
      {
        "id": "5753026921",
        "name": "End-user subscription permission set",
        "href": "https://mycsp.com/tmf-
api/rolesAndPermissionsManagement/v5/PermissionSet/5435648",
        "@type": "PermissionSetRef",
        "@referredType": "PermissionSet"
      }
  ],
  "locale": "en",
```


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```
   "middle_name": "Mary",
   "name": "Jane Mary Doe",
   "nickname": "JaneDoe",
   "phone_number": "+1 (425) 555-1212",
   "phone_number_verified": true,
   "picture": "https://someurl.com/janeDoe.jpg",
   "preferred_username": "janeDoe",
   "profile": "https://my.blog.com/janeDoe",
   "sub": "a066ffd5-f685-43bc-ad57-bfb7639decb0",
   "website": "https://janedoe.com",
   "zoneinfo": "America/Los_Angeles"
   }
 ]
```


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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
1.0	03-Jul-2018	Luis Velarde (Telefónica)	First Release of Draft. Version of the Document.
1.0.1	10-Sep-2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status.
4.0	28-May-2020	Alan Pope	Alignment with Guidelines 4.0.
4.0.1	20-Jul-2020	Adrienne Walcott	Updated to reflect TM Forum Approved Status.
5.0	31-Mar-2024	Dominic Oyeniran dominic.oyeniran@vodafone.com (Vodafone Group)	Convert to v5. Added global party domain overview, user permission set, security scheme, api basepath update etc.

Contributors to Document

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